Arista Access Node Routing Protocol Flap Resolution Procedure for BTM-ACC-19

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8088

Description

The Arista access node, situated at BTM Layout Aggregation, designated as BTM-ACC-19, is experiencing a routing protocol flap, denoted by alarm code ALARM_8088 with a severity of Major. This failure domain is precipitated by the intricacies of the routing protocols in operation, specifically BGP, OSPF, and ISIS, which are pivotal in maintaining the stability and reachability of the network. The BGP route churn, OSPF adjacency loss, and ISIS LSP flooding are indicative of fluctuations in the routing table, potentially triggered by changes in network topology, misconfigurations, or hardware malfunctions. The Arista access node, as a critical component of the network infrastructure, relies on the seamless operation of these protocols to ensure uninterrupted data transmission. Therefore, a comprehensive understanding of the underlying protocol mechanics and the Arista CLI dialect is essential for effective troubleshooting and resolution. The routing protocol flap can lead to network instability, packet loss, and decreased overall performance, necessitating prompt identification and rectification of the root cause.

Pre-requisite Checks

- Verify the physical integrity of the access node and its connections
- Ensure redundant link status is active and functional
- Dispatch remote hands to BTM Layout Aggregation for on-site assistance if necessary
- Confirm the Arista node is running the latest software version

Potential Root Causes

Cause	Probability
BGP Route Churn due to Peering Issues	High
OSPF Adjacency Loss resulting from Network Topology Changes	Medium
ISIS LSP Flooding caused by Hardware Malfunction	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP peering status and route information, Description: This command isolates issues in the BGP control plane by displaying the status of BGP peers, the number of prefixes received, and any errors encountered. It helps in identifying peering issues that could lead to route churn.

Command: show ip ospf neighbor, Expected Output: OSPF neighbor information and adjacency status, Description: This command examines
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the OSPF control plane by listing all OSPF neighbors, their states, and the dead timer values. It aids in diagnosing adjacency losses and network topology changes affecting OSPF.

Command: show isis database, Expected Output: ISIS link-state database information, Description: This command delves into the ISIS control plane, displaying the link-state database that contains information about the network topology. It is useful for identifying LSP flooding issues and understanding the current network topology as seen by ISIS.
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Resolution Steps

Step 1: Step: Adjust BGP Peering Configuration, Command: configure terminal ; router bgp ; neighbor description ; neighbor ebgp-multihop , Description: Modify the BGP peering configuration to stabilize the BGP session, including setting appropriate descriptions for easier management and adjusting the EBGP multihop TTL to ensure proper peering establishment across multiple hops.

Step 2: Step: Restore OSPF Adjacency, Command: configure terminal ; router ospf ; area ; network area , Description: Reestablish OSPF adjacency by ensuring the correct network configuration under the appropriate OSPF area, which may involve adjusting the network statement to match the current topology.

Step 3: Step: Mitigate ISIS LSP Flooding, Command: configure terminal ; router isis ; is-type level-1 ; lsp-refresh-interval , Description: Configure ISIS to reduce LSP flooding by setting the LSP refresh interval to a more appropriate value, which can help in stabilizing the network and reducing the load on the ISIS protocol.

Post-Check Verification

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Command: show ip bgp, Expected Output: Detailed BGP routing table information, Description: Verify that the BGP routing table is stable and reflects the expected routes after adjusting the peering configuration.

Command: show ip ospf interface, Expected Output: Detailed OSPF interface information, Description: Confirm that OSPF interfaces are in the expected state and that adjacencies have been reestablished successfully.

Command: show isis spf-log, Expected Output: ISIS SPF calculation log information, Description: Check the ISIS SPF log to ensure that there are no recent SPF calculations that would indicate ongoing LSP flooding issues.
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Roll-back Procedures

- Step: Revert BGP Configuration, Command: configure terminal ; no router bgp ; no neighbor , Description: If the BGP peering adjustments cause unforeseen issues, revert the BGP configuration to its previous state by removing the newly added commands.

- Step: Restore Original OSPF Configuration, Command: configure terminal ; no router ospf ; no area ; no network area , Description: If changes to the OSPF configuration lead to instability, remove the adjustments made and revert to the original OSPF configuration.

- Step: Revert ISIS Configuration, Command: configure terminal ; no router isis ; no is-type level-1 ; no lsp-refresh-interval , Description: If the ISIS configuration changes exacerbate the LSP flooding, remove the new settings and revert to the previous ISIS configuration to stabilize the network.